TLA+ Study Group — Week 01

Aleksandr

November 23, 2025

Definitions

0.1 What is a TLA+?

- It is the language for modeling digital systems with an included feature of model properties verification. Abstract enough to be above of code- and harware- level
- Suitable for modeling concurrent and distributed systems, allows to find design flaws before code writing.
- In TLA+ models, the execution of a system is represented by a *finite sequence of discrete state changes* - behaviour.

0.2 Abstraction

- It is a simplified model of an object with removed irrelevant in the current problem context details.

0.3 Specification

- The precise high-level model.
- On higher levels can be called "architecture."

0.4 State machine

- is an abstract model representing behaviour of a digital system.
- is defined by specifying (a) - state definition, (b) - ALL initial states, and (c) - ALL possible transitions between states.

0.5 Notations

- For variable x: x' - is the next x value.
- $\{a_0, a_1, ..., a_n\} \longmapsto a_0..a_n$
- commutative operators **and, or** are represented by: $\wedge$, $\vee$
- Upper-case keywords: `TRUE, FALSE, IF, THEN, ELSE`
- "def" $\longmapsto \triangleq$

Example of spec

```
EXTENDS Integers
VARIABLES x, pc

Init == (pc = "start") /\ (x = 0)

Assign ==    /\ pc = "start"
             /\ x' \in 0..100
             /\ pc' = "middle"

Increment ==    /\ pc = "middle"
                /\ x' = x + 1
                /\ pc' = "done"

Next == Assign \/ Increment
```